

Market Data Source, Consistency & Load Audit

NSA / MarketPil intraday trading application — forensic data-flow audit

Written for: the owner and anyone deciding what to fix next · AUDIT ONLY — no code, providers, polling, or logic were changed · Every claim is tagged CODE VERIFIED or NOT VERIFIED (needs runtime telemetry) · Generated 2026-10-02

1. Executive Summary

- **The app is effectively single-provider: Dhan.** `getProvider()` unconditionally returns a `DhanProvider` (all quotes, candles, OI chain, India VIX). The "always Groww" comments in the code are **stale/incorrect**; `growwProviderForOi()` even returns a `DhanProvider`. CODE VERIFIED
- **Groww is used in exactly one place:** Option Terminal / option-structure **premium (LTP) candles** for **NSE options only** (SENSEX/BANKEX/BANK NIFTY are explicitly excluded and use Dhan), plus a token probe on connect and the Groww instrument master (`findOption`). CODE VERIFIED
- **Twelve Data** is used only by the Market Sentiment desk for external global/macro/FX/commodity data (needs `MARKET_DATA_API_KEY`; otherwise DATA UNAVAILABLE). `FII_DATA_URL` is an optional FII source. No Yahoo/TrueData/NSE/mock for live Indian data. CODE VERIFIED
- **The architecture is already well-cached and de-duplicated.** A shared in-memory `_cache` in `api.ts` plus `getOiCached` mean screens do not each hit Dhan independently — results are shared by symbol. The 5-second frontend polls mostly hit cache; real Dhan calls are gated by TTLs (OI 90s, candles 20s). CODE VERIFIED
- **The genuine risks are narrow:** (1) `/api/quote/:symbol` is **uncached**; (2) the OI-Analysis spot-change quote is **uncached**; (3) Option Terminal **mixes Dhan OI/spot with Groww premium candles**; (4) the **"1 Min" OI cadence is really ~90s** (bounded by `TTL_OI`), not 60s.

2. Screen → Data Source Map (CODE VERIFIED)

SCREEN	ROUTE(S) HIT	PRIMARY	SECONDARY	POLL (OPEN)	PANEL-GATED
Market Command	<code>/api/market-command</code>	Dhan	—	5s	yes
Option Terminal	market-command, option-candles, option-structure	Dhan (spot/OI)	Groww (NSE option candles)	5s	yes
Trade Execution	market-command, trade-log, orb-strategy, vwapema-strategy	Dhan	—	5s	yes
OI Analysis (Movement/Summary)	oi-analysis/movement · summary · strike-history	Dhan	—	30s	yes
Market Sentiment (premarket)	premarket/overview · health	Dhan (Indian)	Twelve Data, FII_DATA_URL	20s (srv-cached 30s)	yes
OI Command	oi-command (buildOiCommand)	Dhan	—	~5–15s	yes
Liquidity Status	liquidity-status	Dhan	—	—	yes
Index strip (Option desk)	<code>/api/quote/:symbol</code> x4	Dhan	—	5s	mode-option only
Global status bar	data-status, best-trade, top-opportunities	Dhan	—	15s/60s/60s	always-on
Watchlist / Index desk	quotes + scan	Dhan	—	60s	always-on

Decision Flow, AI/Paper, Backtest, Swing/Stock screens: CODE VERIFIED to run off the same Dhan provider + shared cache; detailed per-screen call counts NOT VERIFIED (outside the live-market critical path).

3. Dhan Data Usage (CODE VERIFIED)

- **Quotes/LTP:** `/api/quote/:symbol` (uncached), market-command spot, sentiment quotes, OI-Analysis spot-change.
- **Candles:** `fetchCandles / getCandlesCached` → `c:${symbol}:${interval}` cache.
- **Option chain / OI / Greeks / OI-change:** `getOiCached` → `dhanOiAnalysis` → `oi:${symbol}` cache.
- **India VIX:** `getIndiaVix` (`india-vix` cache 15s). **SENSEX / BANK NIFTY option candles:** Dhan directly.

4. Groww Data Usage (CODE VERIFIED)

- `api.ts:7010–7022` `/api/option-candles` → `growwProviderForOptionTerminal()` + `growwOptionCandles()` (NSE premium candles).
- `api.ts:5200–5203` `/api/option-structure` → same, for the S-R premium series.
- `api.ts:6146` `GrowwProvider(token)` — token validation on `/groww/connect`.
- `growwInstruments.findOption/optionExpiries/optionStrikes` — Groww instrument master.
- Guard: `growwSupports = !SENSEX|BANKEX|BANKNIFTY|i.test(underlying)` — those go straight to Dhan.

5. Other Sources

- **Twelve Data** — Market Sentiment external segments only; UNAVAILABLE without `MARKET_DATA_API_KEY`. CODE VERIFIED
- `FII_DATA_URL` — optional FII futures JSON; unset ⇒ DATA UNAVAILABLE. **RSS news** — Sentiment/news screens.
- No Yahoo / TrueData / NSE / mock in the live path. CODE VERIFIED

6. Dhan/Groww Mixing (CODE VERIFIED that the mix exists)

Only the Option Terminal mixes providers: spot/index → Dhan (market-command); option OI / S-R walls / Greeks → Dhan (`getOiCached`); option premium (LTP) candles on the CE/PE chart → Groww (NSE) / Dhan (BSE + BANKNIFTY).

7 & 8. Data Inconsistencies

SCREEN	FIELD A	FIELD B	ROOT CAUSE	SEVERITY	USER IMPACT
Option Terminal	Dhan OI / S-R levels	Groww premium candles	Separate feeds + separate cache cycles (OI 90s vs Groww fetch)	MEDIUM	S-R lines from Dhan OI may be slightly time-misaligned vs the Groww premium chart
OI Analysis	spot / spotChg (uncached Dhan quote)	OI chain underlying (90s cache)	Two Dhan reads at different times	LOW	Spot in the strip can differ slightly from the chain's underlying
Any screen (market hours)	live quote	OI chain	Different TTLs (quote fresh vs OI ≤90s)	LOW	Minor; both Dhan, both freshness-labeled

Exact value/timestamp deltas: **NOT VERIFIED — requires runtime telemetry**.

9. Top Load Sources (CODE VERIFIED cadence; network rate NOT VERIFIED)

The shared cache means most frontend polls hit cache, so HTTP-poll rate ≠ Dhan-call rate.

#	OPERATION	CADENCE	FAN-OUT	CACHED?	REAL DHAN PRESSURE
1	warmCoreFeeds (background)	45s (feed window)	5 idx × (5m+15m candles + OI) + 6 eq 15m ≈ 21 calls	writes shared cache	Highest steady
2	/api/quote/:symbol (index strip)	5s	×4 symbols	NO	High if mode-option active (likely dormant)
3	Active screen /api/market-command (MC/OT/TE)	5s	1 symbol → candles+OI+vix+ls	YES (15–90s)	Low (cache hits); refills on TTL expiry
4	OI-Analysis spot-change quote	30s	×1 movement + ×1 summary	NO	Low but uncached
5	evaluateOiSignals (background)	5min	getQuote per symbol	partial	Low
6	refreshOiChangeSnapshot	3min	OI per symbol	shares oi:	Low
7	Option Terminal /api/option-candles	5s	Groww (NSE)	short cache	Groww load, not Dhan
8	best-trade / top-opportunities / data-status	60/60/15s	1 each	best-trade 60s	Low

Ranks 9–20 (paper tick ~5min, OI scalp 90s, trade monitor 60s, ORB auto-log 60s, WhatsApp 90s, dhan-health 60s, syncSessionProvider 30s, credential rotate 60s, OI sampler, feed-heal) are all low / non-Dhan / cache-shared. A precise Top-20 by requests/min is **NOT VERIFIED**.

10. Duplicate Requests (CODE VERIFIED)

The obvious duplication is already prevented: OI Movement, OI Summary, Market Command, Trade Execution and OI Command all read the same oi:\${symbol} cache (getOiCached, TTL 90s + in-flight coalescing). They do **not** each hit Dhan for the chain.

Remaining genuine duplication: (1) OI-Analysis spot quote duplicates the chain underlying (2 uncached getQuote / 30s); (2) /api/quote/:symbol uncached re-fetch every 5s with no coalescing (only when mode-option); (3) Option Terminal Groww premium + Dhan OI fetched separately for the same visual (by design).

11. Polling Audit (CODE VERIFIED)

TIMER	INTERVAL	ENDPOINT	MARKET-GATED	PANEL-GATED	NOTE
MC/OT/TE tick	5s (30s closed)	/api/market-command	yes	yes	only active panel
OI Analysis	30s	oi-analysis/*	active-check	yes	
Premarket	20s	premarket/overview	active-check	yes	server-cached 30s
index strip	5s	/api/quote ×4	mode-option + lmc-fullwidth	—	uncached; likely dormant desk
dhan-config	5s	/api/groww/config	no	—	config, not market data
data-status	15s	/api/data-status	no	—	
best-trade	60s	/api/best-trade	no	—	cached 60s
top-opportunities	60s	/api/top-opportunities	yes	lmc-fullwidth	
MC index cards	15s	none (renders MC.lastData)	—	yes	no fetch
smart news	5min	news	—	—	

Most aggressive: the 5s MC/OT/TE poll (cache-served) and the 5s index strip (uncached, gated to a likely-dormant desk). /api/groww/config at 5s is config polling, unnecessarily frequent.

12. Option-Chain Load (CODE VERIFIED)

- Full chain fetched per symbol via dhanOiAnalysis, cached as oi:\${symbol} (90s, topStrikes = ±12/25 strikes, single expiry). One fetch feeds all OI consumers.
- Calls and puts arrive in one response (not separate). No per-strike fetching.
- Chain refresh ≈ once per 90s per symbol (TTL_OI) or on warmCoreFeeds (45s, but also hits the 90s cache).

13. OI Screen Audit (CODE VERIFIED)

- Source:** Dhan only (getOiCached), shared across OI Movement/Summary/Command/MC/TE.
- ΔOI:** computed **locally** — day ΔOI from the chain (ceChg/peChg); intraday ΔOI from the in-memory intradayStore (per-strike snapshots).
- Heatmap 1/3/5/15/30-min:** local windowed diffs; falls back to day-change (dimmed) until a window fills.

- **"1 Min" reality — FINDING:** snapshot cadence is bounded by `TTL_OI = 90s` (chain asOf changes ~every 90s) plus `MIN_SAMPLE_GAP_SEC = 55s`. So "1 Min" Δ OI is effectively **~90-second** granularity, not a true 60-second snapshot.
- Chart uses **absolute OI** for strike bars and **Δ OI** for the change/heatmap panels.

14. Timestamp / Staleness Audit

Quotes carry the provider `marketTime`; freshness (LIVE/DELAYED/STALE/CLOSED) is derived from it (recently hardened so cached quotes recompute freshness). OI chain carries `asOf` + an age badge. **CODE VERIFIED**. Actual live ages per field: **NOT VERIFIED — needs runtime telemetry**.

15. Cache Audit (CODE VERIFIED)

CACHE KEY	TTL	SHARED BY
<code>oi:\${symbol}</code>	90s (+90s last-good)	all OI screens, MC, TE
<code>c:\${symbol}:\${interval}</code>	20s intraday / 60s (60m) / 10min (1d)	MC, OT, TE, warm
<code>oi-command, ls, india-vix, oi-command-ext</code>	15s	MC, OI Command
<code>premarket:overview</code>	30s	Market Sentiment
<code>sentiment _qCache</code>	25s open / 30min closed	Market Sentiment quotes
<code>best-trade</code>	60s	status bar
<code>/api/quote/:symbol</code>	none	index strip
<code>OI-Analysis spotChg quote</code>	none	OI Analysis

16. Market-Hour Performance Risk

RISK	SEVERITY
<code>/api/quote/:symbol</code> uncached, 5s x4 (if mode-option active)	MEDIUM
OI-Analysis spotChg quote uncached (2x / 30s)	LOW
Option Terminal Dhan-OI ↔ Groww-premium time misalignment	MEDIUM
"1 Min" OI label vs ~90s real cadence (mislabel, not overload)	MEDIUM (correctness)
warmCoreFeeds 45s burst competing with Dhan rate limit (DH-904 seen this session)	MEDIUM
<code>/api/groww/config</code> polled every 5s	LOW

17–19. Findings by Priority

CRITICAL: none found in code (no unbounded per-strike fetching, no live fabrication, OI chain is shared).

HIGH: none purely from code; the Option Terminal provider-mix alignment could be HIGH *if* runtime shows large deltas — **NOT VERIFIED**.

MEDIUM: (a) `/api/quote` uncached; (b) Option Terminal Dhan/Groww alignment; (c) "1 Min" OI is ~90s; (d) warmCoreFeeds burst vs Dhan rate limit.

LOW: OI-Analysis uncached spot quote; 5s groww/config poll; index-strip gated to a possibly-dead mode-option desk.

NOT VERIFIED (needs runtime telemetry): exact requests/min & /sec, concurrent-request counts, payload/response sizes, real Dhan-call rate vs cache-hit ratio, actual Dhan↔Groww value/timestamp deltas, true Top-20 ranking, per-screen latency during live hours.

20. Master Table

SCREEN	DATA TYPE	PROVIDER	ENDPOINT/SERVICE	POLL	CALLS/MIN (POLL)	CACHE	DUPLICATE	TIMESTAMP	CONSISTENCY	LOAD RISK
Market Command	spot/candles/OI	Dhan	market-command	5s	~12	yes 15–90s	shares oi:	marketTime/asOf	consistent	LOW
Option Terminal	spot/OI (Dhan) + premium (Groww)	Dhan+Groww	market-command, option-candles, option-structure	5s	~12	partial	Groww separate	mixed feeds	mixed	MEDIUM
Trade Execution	spot/candles/OI + strategy	Dhan	market-command, trade-log, orb/vwapema	5s	~12	yes	shares oi:	Dhan	consistent	LOW
OI Analysis	OI chain + spot	Dhan	oi-analysis/movement-summary-strike-history	30s	~4	yes (oi:) + uncached spot	shares oi:	asOf + uncached spot	mostly consistent	LOW–MED
Market Sentiment	Indian (Dhan) + global (TwelveData) + FII	Dhan+TwelveData	premarket/overview-health	20s	~3	yes 30s + 25s/30min	no	provider ts + CLOSED	consistent per source	LOW
Index strip	quotes	Dhan	<code>/api/quote/:symbol</code> x4	5s	~48	NO	yes	marketTime	consistent	MEDIUM (if active)
Status bar	signal/status	Dhan	best-trade, data-status, top-opps	15–60s	~4	best-trade 60s	no	mixed	consistent	LOW

21. Final Answers

A. DHAN only: Market Command, Trade Execution, OI Analysis (Movement/Summary/Heatmap), OI Command, Liquidity Status, Index strip, Watchlist, status bar, Decision Flow, Backtest, most Swing/Stock/AI screens.

B. GROWW only: none (Groww is never a whole screen).

C. BOTH Dhan + Groww: Option Terminal (Dhan spot/OI + Groww NSE premium candles) — the only one.

D. Inconsistent-risk screens: Option Terminal (Dhan OI vs Groww premium timing); OI Analysis (uncached spot vs chain underlying) — minor.

E. Highest market-hour load: warmCoreFeeds (45s background fan-out) and the uncached `/api/quote` index strip (5s×4) if that desk is active; then the active MC/OT/TE 5s poll (mostly cache-served).

F. Top duplicate calls: OI-Analysis spot quote vs chain underlying; uncached `/api/quote` re-fetches; (everything else is de-duped by the shared cache).

G. Top polling issues: 5s MC/OT/TE (aggressive but cached); 5s uncached index strip; 5s groww/config; 20s premarket (fine, server-cached).

H. Top architecture risks: uncached `/api/quote`; Dhan/Groww mix in Option Terminal; "1 Min" OI actually ~90s; warmCoreFeeds burst vs Dhan DH-904; stale "always Groww" comments misleading future devs.

I. What to fix FIRST (proposals only — awaiting approval; no code changed):

1. **Cache `/api/quote/:symbol`** (short TTL + in-flight coalescing) — removes the only uncached hot path.
2. **Reuse the chain underlying** for OI-Analysis spot (drop the 2 uncached quotes).
3. **Relabel/clarify the "1 Min" OI** to reflect ~90s cadence, or lower `TTL_OI` intraday to genuinely support 1-min (trade-off: more Dhan calls).
4. **Align Option Terminal's Dhan-OI and Groww-premium timestamps** (or annotate the S-R overlay with its own asOf).
5. **Fix the stale "always Groww" comments** so the Dhan-only reality is clear.

18. Proposed AUDIT MODE (proposal only — not implemented)

A feature-flagged (`AUDIT_MODE=false`) request-logger wrapping the Dhan/Groww clients, appending one JSONL line per outbound call: `{ts, screen(referer), session, provider, endpoint, fn, reqId, instrument, securityId, expiry, strike, optType, tStart, tEnd, latencyMs, httpStatus, respBytes, cacheHitMiss, retry, error, pollSource}`, plus a `/api/audit/summary` rollup (calls/min per endpoint, cache-hit %, duplicate %, p95 latency). This converts every NOT VERIFIED item above into hard numbers. Not built — awaiting go-ahead.

Audit only — no code, providers, polling, cache, UI, database, or trading behavior was changed. The system is Dhan-centric and already well-cached/de-duplicated; the fixable items are the handful of uncached hot paths, the Option Terminal provider mix, and the "1 Min" OI cadence label.